Representations of finite groups

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Contents

1 An overview of representation theory

$$\begin{tikzcd}
 & \mathcal{C} = \textbf{Set} : \text{Set-theoretic representation} \\
 & \mathcal{C} = \textbf{Vsp}_{\mathbb{K}} : \text{Linear representation} \\
\text{Functor } F : \textbf{Gp} \to \mathcal{C} \arrow[uur] \arrow[ur] \arrow[r] \arrow[dr] \arrow[ddr] & \mathcal{C} = \textbf{AbGp} \\
 & \mathcal{C} = \textbf{Pro} : \text{Projective representation} \\
 & \mathcal{C} = \textbf{Aff} : \text{Affine representation} \\
 & \mathcal{C} = \textbf{Top}
\end{tikzcd}$$

Another perspective of representation functor: It is just group actions on different sets, so representations could be viewed as a continuation of group actions:

2 Basic definition of linear representations

2.1 Preliminary settings

2.2 Categorical stuff

A finite group G can be viewed as a category, denoted $\mathbf{B}G$, who has single object $*$ and $\mathrm{Hom}(*,*) =: G$, the collection of morphisms is defined to be the group elements $g : * \to *$. The group operation perfectly matches the composition rules of morphisms. Hence, compositions are group multiplications and identity is the identity element in group G. All axioms of being a category is satisfied.

Proposition 2.1: Linear representations as functors

Let G be a finite group an V be a vector space over a field $\mathbb{F}$. Then, a linear representation (ρ, V) is a functor

$$\mathbf{B}G \to \mathbf{fdVect}_{\mathbb{F}}$$

Proposition 2.2: Category of linear representations of finite group over a field

Let G be a finite group and $\mathbb{F}$ be a field. All linear representation of G over $\mathbb{F}$ forms a category, denoted as $\mathbf{Rep}_{\mathbb{F}}(G)$:

- Objects: Pairs (ρ, V). For each (ρ, V), V is a vector space over $\mathbb{F}$ and ρ is a linear representation of G on V.
- Morphisms: Equivariant maps.
- Compositions: The composition of equivariant maps.
- Identities: Identity equivariant maps.

Remark From the perspective provided in Proposition 2.1, objects of $\mathbf{Rep}_{\mathbb{F}}(G)$ are the functors $\mathbf{B}G \to \mathbf{fdVect}_{\mathbb{F}}$ and the morphisms, G-equivariant maps, are natural transformations.

Corollary 2.1: $\mathbf{Rep}_{\mathbb{F}}(G)$ as a functor category

2.2.1 Group Ring languages

Consider a representation $\rho : G \to GL(V)$. For V, as a linear space, it includes the scalar multiplication action of the base field $\mathbb{C}$ on V; simultaneously, as a representation, V also includes the linear transformation action of the group G on V. These two actions are mutually respectful and absolutely compatible. This means:

$$\rho(g)(c \cdot v) = c \cdot (\rho(g)(v))$$

That is, whether the scalar from $\mathbb{C}$ acts first or the group element g acts first, the result remains strictly consistent. However, viewing these two actions in isolation brings about a fragmentation in the algebraic structure—there is a lack of unified connection between the multiplication of the group G and the multiplication of $\mathbb{C}$ at a general algebraic level. In order to extend the group action smoothly and linearly, we consider expanding the base field $\mathbb{C}$ into a unified entity that contains both the group structure of G and the field structure of $\mathbb{C}$: the group ring $\mathbb{C}[G]$. Consequently, V naturally becomes a $\mathbb{C}[G]$-module. Treating V as a $\mathbb{C}[G]$-module perfectly encompasses all the underlying basis structures of V acting as a linear space over $\mathbb{C}$. Once this step is completed, representation theory is no longer a geometric process of studying "how a group dynamically flips within a linear space". It is strictly packaged into a pure, static algebraic problem: studying the module structure over a specific group ring $\mathbb{C}[G]$. This dimensional reduction in perspective corresponds to an absolute equivalence of categories:

$$\mathbf{Rep}_{\mathbb{F}}(G) \cong_{\mathbb{F}[G]} \mathbf{Mod}$$

Under this equivalence, all the geometric language in representation theory can be accurately translated into the language of module: A G-invariant representation subspace is translated into a purely algebraic submodule. A G-equivariant map that preserves the group action is translated into a module homomorphism that preserves the ring action.

When representation theory is established on an algebraically closed field of characteristic 0, such as $\mathbb{C}$, the linear structure obeys the rule of semisimplicity, and representations can be completely decomposed into a direct sum of irreducible representations. However, this property is highly dependent on the choice of the base field. If we switch the base field to the field of rational numbers $\mathbb{Q}$, or to a finite field $\mathbb{F}_p$ of characteristic p, this semisimple direct sum decomposition will no longer universally hold, and its underlying algebraic structure will subsequently become much more complex.

not only semisimplicity, group rings remains...

2.2.2 Equivalent class of representation

Classify representation of G up to isomorphism: *Isomorphism class* is a collection of linear representations of G such that any two representations in that collection are isomorphic (in the sense of representation).

Lemma 2.0.1 *Any element of an isomorphism class can be represented by* $(\tau, \mathbb{C}^n)$

2.3 New representations from old ones

2.3.1 Overview of new representations

direct sum $(\rho_1 \oplus \rho_2, V_1 \oplus V_2)$

tensor product $(\rho_1 \otimes \rho_2, V_1 \otimes V_2)$

Given $(\rho_1, V_1), (\rho_2, V_2)$ induced

Hom $(\sigma, \mathrm{Hom}(V_1, V_2))$

2.3.2 Direct sum of representations

2.3.3 Tensor product of representations

Definition 2.1: Tensor product of representations

Let (ρ_i, V_i) be representations of a group G. The **tensor product representation**, denoted $(\rho_1 \otimes \rho_2, V_1 \otimes V_2)$, has an underlying vector space $V_1 \otimes V_2$, given by the homomorphism $\rho_1 \otimes \rho_2 : G \to \mathrm{GL}(V_1 \otimes V_2)$ given by

$$\rho_1 \otimes \rho_2(g) := \rho_1(g) \otimes \rho_2(g)$$

where $\rho_1(g) \otimes \rho_2(g)$ is the tensor product of linear maps.
More explicitly,

$$\rho_1 \otimes \rho_2(g)(v_1 \otimes v_2) = \rho_1(g)(v_1) \otimes \rho_2(g)(v_2)$$

or

$$\rho_1 \otimes \rho_2(g) \left(\sum_{i=1}^{n} v_i \otimes w_i \right) = \sum_{i=1}^{n} \rho_1(g)(v_i) \otimes \rho_2(g)(w_i)$$

2.4 Characters of representation

Definition 2.2: Character of representation

Given a linear representation $\rho : G \to \mathrm{GL}(V)$, its character χ_ρ is defined as

$$\chi_\rho : G \to \mathbb{C} \quad \chi_\rho(g) = \mathrm{Tr}(\rho(g)) \text{ or } g \mapsto \mathrm{Tr}(\rho(g))$$

Proposition 2.3: Properties of characters

Let (ρ, V) be a linear representation of G and χ_ρ is the chatacter of ρ. Then:
(1) χ_ρ only depends on the isomorphism class of ρ.
(2) χ_ρ is constant on each conjugacy class.
(3) $\chi_\rho(1_G) = \dim V$

Proof: (3) $\chi_\rho(1_G) = \mathrm{Tr}(\rho(1)) = \mathrm{Tr}(\mathrm{Id}_V) = \dim V$ □

Theorem 2.1: Identification of character group

$\exists$ a natural isomorphism

$$G^* \cong (G^{ab})^*$$

where $G^{ab} = G/G'$ and G' is the commutator subgroup of G.

Proof: Using the Universal property of abelianization, $\forall$ morphism $f : G \to \mathbb{C}^\times$, $\exists!$ morphism $g : G^{ab} \to \mathbb{C}^\times$ such that $g \circ \pi = f$. This proves the surjectivity of $(G^{ab})^* \to G^*$
injectivity □
So abelian group is isomorphic to its character group.

2.5 Special representations for specific groups

2.6 Decomposition of representations (as direct sum)

Definition 2.3: Subrepresentation

Let (ρ, V) be a representation of G and $U \leq V$ such that

$$\forall g \in G,\ \forall u \in U,\ \rho(g)(u) \in U$$

Then $(\rho|_U, U)$ is a subrepresentation of (ρ, V), where

$$\rho|_U : G \to \mathrm{GL}(U), \quad \text{by } g \mapsto \rho(g)|_U$$

Namely, $\rho|_U(g) := \rho(g)|_U$

2.6.1 An important subrepresentation: Projection

$$\begin{tikzcd}
(\rho, V) \arrow[r, leftrightarrow, "\pi\ [2]"] & (\rho|_{V^G}, V^G) \arrow[l, bend left, "\geq\ [1]"] \arrow[r, dashed, "{[3]}"] & \dim V^G
\end{tikzcd}$$

2.7 Maschke's theorem

2.8 A special decomposition

2.8.1 Dual representation

2.8.2 Structure of Hom representations

$$\begin{tikzcd}
(\rho, V) \arrow[dr, "{[1]}"'] & (\tau, W) \arrow[d, "{[1]}"] & \\
& (\sigma, \mathrm{Hom}(V, W)) \arrow[r, equal, "\sim\dagger\ {[3]}"] \arrow[d, "{[2]}"', "{(\tau,W)=(\mathbb{1},\mathbb{C})}"] & (\rho^\vee \otimes \tau, V^\vee \otimes W) \\
& (\rho^\vee, V^\vee) &
\end{tikzcd}$$

What is the relationship between $\mathrm{Hom}_G(V, W)$ and $\mathrm{Hom}(V, W)$? Illustrated by the following theorem, it is simply

$$\mathrm{Hom}_G(V, W) = \mathrm{Hom}(V, W)^G$$

Theorem 2.2: Identification of Hom

Let (ρ, V) and (τ, W) be two linear representations. $\mathrm{Hom}_G(V, W) = \mathrm{Hom}(V, W)^G$

Proof:

$$\begin{aligned}
\mathrm{Hom}(V, W)^G &= \{T : V \to W | \forall g \in G,\ \sigma(g)(T) = T\} \\
&= \left\{T : V \to W | \forall g \in G,\ \tau(g) \circ T \circ \rho(g)^{-1} = T\right\} \\
&= \{T : V \to W | \forall g \in G,\ \tau(g) \circ T = T \circ \rho(g)\} \\
&= \mathrm{Hom}_G(V, W)
\end{aligned}$$

$\square$

Proposition 2.4: Character of $(\sigma, \mathrm{Hom}(V, W))$

Let (ρ, V) and (τ, W) be two linear representations. The character of $(\sigma, \mathrm{Hom}(V, W))$ is given by

$$\chi_\sigma = \chi_\tau \cdot \overline{\chi}_\rho$$

Lemma 2.1: Schur's lemma

Let (ρ, V) and (τ, W) be two irreducible linear representations of G. Then

$$\mathrm{Hom}_G(V, W) \cong \begin{cases} \mathbb{C} & (\rho, V) \cong (\tau, W) \\ 0 & \text{otherwise} \end{cases}$$

2.9 Class functions

struture of $\mathrm{Class}(G)$

$$\begin{tikzcd}
\text{Space of class functions } \mathrm{Class}(G) \arrow[r, "{[1]}"] \arrow[dr, "{[3]}"'] & \text{a vector space} \arrow[r, "{[2]}"] & \text{basis} \\
& \text{a Hermitian space} &
\end{tikzcd}$$

Definition 2.4: Class function

A function $f : G \to \mathbb{C}$ is a class function if

$$\forall g, h \in G,\ f(hgh^{-1}) = f(g)$$

i.e. f is constant on each conjugacy class.
The **space of class function**, denoted $\mathrm{Class}(G)$, is $\mathrm{Class}(G) := \{f : G \to \mathbb{C} | f \text{ is a class function}\}$.

3 Induced representation

3.1 Induced representations (first encounter)

3.2 Induced representations (second encounter)

As in the first encounter, let G be a finite group with a subgroup H. Let (ρ, W) be a linear representation of H. The representation of G induced by W is denoted $({}^G_H\rho, {}^G_H W)$.

3.2.1 Characters of induced representations

3.2.2 Restriction to subgroups

Let G be a finite group with subgroups $H, K \leq G$ and (ρ, W) be a linear representation of H. Instead of inducing a representation from H to G and restricting it back to H, ${}^G_H W$ is restricted to another subgroup K.
Here we need double cosets: [Dun23], [Ser77b]

Definition 3.1: Mackey's decomposition

Let G, H, K, S, W have the same meaning as in the prescribed context.
The representation $(\mathrm{res}^G_K{}^G_H\rho, \mathrm{res}^G_K{}^G_H W)$ can be decomposed into direct sum of $({}^G_{H_s}\rho, {}^G_{H_s} W)$,

$$\mathrm{res}^G_K{}^G_H W \cong \bigoplus_{s \in S \cong KG/H}^{K} {}_{H_s}W_s$$

4 Representations on abelian groups

5 Character table of finite groups

5.0.1 Theory behind the character tables

5.0.2 Some categorical packages

As there is a correspondence between the conjugate classes of the group G and the isomorphism classes of irreducible representations. In 2.2, G is reframed in the category $\mathbf{B}G$, so conjugate classes of G are conjugate classes of morphisms of $\mathbf{B}G$. While representations are objects in $\mathbf{Rep}_{\mathbb{F}}(G)$.

5.1 The cyclic group

5.1.1 C_n

5.1.2 C_∞

5.2 The Dihedral group

5.2.1 D_4

5.2.2 D_n, n even, ≥ 2

Suppose $n = 2k$. $D_n = \langle r, s | \, r^n = s^2 = \mathrm{id}, rsr = r^{-1}\rangle$, the conjugate classes of D_n are $\{\mathrm{id}\}, \dots, \{r^i, r^{n-i}\}, \dots, \{r^k\}$ and $\{sr^{2i}\}, \{sr^{2i+1}\}$.

1-dimensional irreducible representations

For a 1-dimensional irreducible representation, the representation space is simply the 1-dimensional complex space $\mathbb{C}$. In this case, ρ is completely equivalent to its character χ, meaning $\chi(g) \in \mathbb{C}^*$. Since complex multiplication is commutative, the relation $srs = r^{-1}$ of D_{2k} collapses:

$$\chi(srs) = \chi(s)\chi(r)\chi(s) = \chi(s)^2\chi(r) = \chi(r)$$

On the other hand, homomorphism simplifies the same relation:

$$\chi(srs) = \chi(r^{-1}) = \chi(r)^{-1}$$

Equating the two yields

$$\chi(r) = \chi(r)^{-1} \implies \chi(r)^2 = 1$$

Simultaneously, the reflection generator satisfies $s^2 = 1$, it naturally follows that:

$$\chi(s)^2 = \chi(1) = 1$$

Thereby, as a 1-dim representation, the character $\chi(r)$ and $\chi(s)$ of the generators r and s are locked into $\{1, -1\}$.

Then, their free combinations yield exactly 4 valid 1-dimensional irreducible representations:

χ \ $g \in D_n$	r	s
χ_1	1	1
χ_2	-1	1
χ_3	1	-1
χ_4	-1	-1

Then, these cases can be homomorphically extended to all conjugate classes:

5.2.3 D_n, n odd

Consider the group presentation $D_n = \langle x, y | y^2 = x^n = xyxy = 1\rangle$

5.2.4 D_{nh}

5.2.5 D_{∞}

5.2.6 $D_{\infty h}$

5.3 Alternating group

5.3.1 A_4

5.4 Symmetric group

5.4.1 S_4

6 p-group representation

6.1 p-group notions revisited

- Solvable groups
- Supersolvable groups
- Nilpotent groups

The next lemma comes from [Ser77a]

Lemma 6.1: Property of fixed set of a p-group

Let G be a p-group and G acting on a finite set X. X^G is the set of elements of X fixed by G. Then,

$$|X| \equiv |X^G| \mod p$$

Theorem 6.1: Existence of fixed elements in p-group representation

Let V be a vector space over a field of characteristic $p > 0$ and G be a p-group. Let $\rho : G \to \mathrm{GL(V)}$ be a linear representation of G in V. V^G is the fixed set of V by G, where the action $G \curvearrowright V$ is induced by ρ. Then, $V^G \neq \{0\}$, i.e. $\exists$ non-zero element of V fixed by all $\rho(s), s \in G$

Proof: Pick an arbitrary $v \in V \backslash \{0\}$. Define the set X to be

$$X := \langle \rho(s)(v) | s \in G \rangle = \mathrm{span}\{\rho(s)(v)|s \in G\} \subseteq \mathrm{V}$$

- X is an n-dimensional vector space for some n by definition of X. Hence, $X \cong \mathbb{F}_{p^k}^n$. and $|X| = p^m$ for some m.
- Applying lemma 6.1, we have $|X^G| \equiv |X| = p^m \equiv 0 \mod p$, but since $0 \in X^G$, the minimal possibility is $|V^G| \geq |X^G| \geq p$. Therefore, $X^G \neq \{0\}$. □

Theorem 6.2: Irreducible representations of p-group

For any p-group G and any linear representation (ρ, V) of G, where V is a vector space over a field of characteristic $p > 0$. The only irreducible representation of (ρ, V) is the trivial representation.

6.2 Theory of Burnside and Blichfeldt

Theorem 6.3: Burnside's theorem

Let p, q be distinct primes and a, b be non-negative integers. Any group G of order $|G| = p^a q^b$ is solvable.

The next theorem asserts that for a supersolvable groups, in particular, for p-groups, every irreducible representation is induced from a 1-dimensional representation. (c.f. [Gor23])

Theorem 6.4: Blichfeldt's theorem

Let G be a supersolvable group and (ρ, V) be an irreducible representation of G, then $\exists$ a subgroup $J \leq G$ and an 1-dimensional representation ψ of J that

$$(\rho, V) \cong ({}^G_J\psi, {}^G_J\mathbb{C})$$

Proof: $\square$

6.3 Brauer's theorem

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